Computer Organization (CSD206): Tutorial 4

For this tutorial, you will solve questions based on Computer Arithmetic.

Submitting this Assignment: You will submit a hardcopy of the solved assignment.

- On the top right side of paper, write your name & roll number and on the top left side, write the date.
- Solve all the questions. Follow the same serial order as given in this sheet.
- This assignment has 0 points (with weightage of 0% in your overall 100 points). **<u>However, you are required to submit it by the due date.</u>**

<u>Questions</u>

1. Represent the following decimal numbers in both sign magnitude and twos complement using 16 bits:
   a) +512
   b) -29
2. Represent the following twos complement values in decimal:
   a) 1101011
   b) 0101101
3. Subtract 101011001 from 110101101011
4. Assume numbers are represented in 4-bit twos complement representation. Show the calculation of the following and indicate overflow (if any):
   a) 6 + 5
   b) -6 + 5
   c) 6 – 5
   d) -6 - 5
5. Use unsigned binary multiplication to multiply 10(multiplicand) by 19(multiplier), where each number is represented using 5 bits.
6. Use the Booth algorithm to multiply -23 (multiplicand) by 29 (multiplier), where each number is represented using 6 bits.